IT PROFESSIONAL PORTFOLIO

Windows Infrastructure and Administration Projects

PROJECT 03

Organizational Unit Design

Windows Server 2025 | Active Directory | OU Architecture | Change Control

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ENVIRONMENT	Personal Windows Enterprise Lab
VERSION	v1.0.0
RELEASE DATE	August 6, 2026
STATUS	Complete

Portfolio documentation standard v1.0.1

Executive Summary

Designed and implemented a scalable Active Directory organizational unit hierarchy for the **corp.mslab.test** forest. The project created a protected **Corp** parent OU, separated identities, endpoints, servers, service accounts, and stale objects into purpose-built containers, and validated accidental-deletion protection through Active Directory Users and Computers and PowerShell.

Business Problem

A flat directory makes policy targeting, delegation, lifecycle management, auditing, and troubleshooting difficult. Before users and computers are onboarded, the domain needs a logical structure that separates object types and business functions while allowing future Group Policy and delegated administration to be applied at the correct scope.

Objectives

- Create a rollback checkpoint before modifying the directory.
- Establish Corp as the administrative parent for custom domain objects.
- Separate admin accounts, groups, servers, service accounts, users, and stale objects.
- Separate desktops and laptops, with Corporate and Branch location scopes beneath each.
- Separate user objects into Finance, HR, Information Technology, and Operations.
- Enable and validate protection from accidental deletion.

Environment

Component	Configuration
Forest / domain	corp.mslab.test; NetBIOS CORP
Domain controller	CORP-DC1 - Windows Server 2025
Administration	Active Directory Users and Computers; Windows PowerShell
Parent OU	OU=Corp,DC=corp,DC=mslab,DC=test
Design size	18 custom OUs total: Corp plus 17 descendant OUs
Safety control	Pre-change checkpoint reported; deletion protection validated
Scope boundary	No users, groups, computer moves, delegation, or GPO links created

OU Architecture

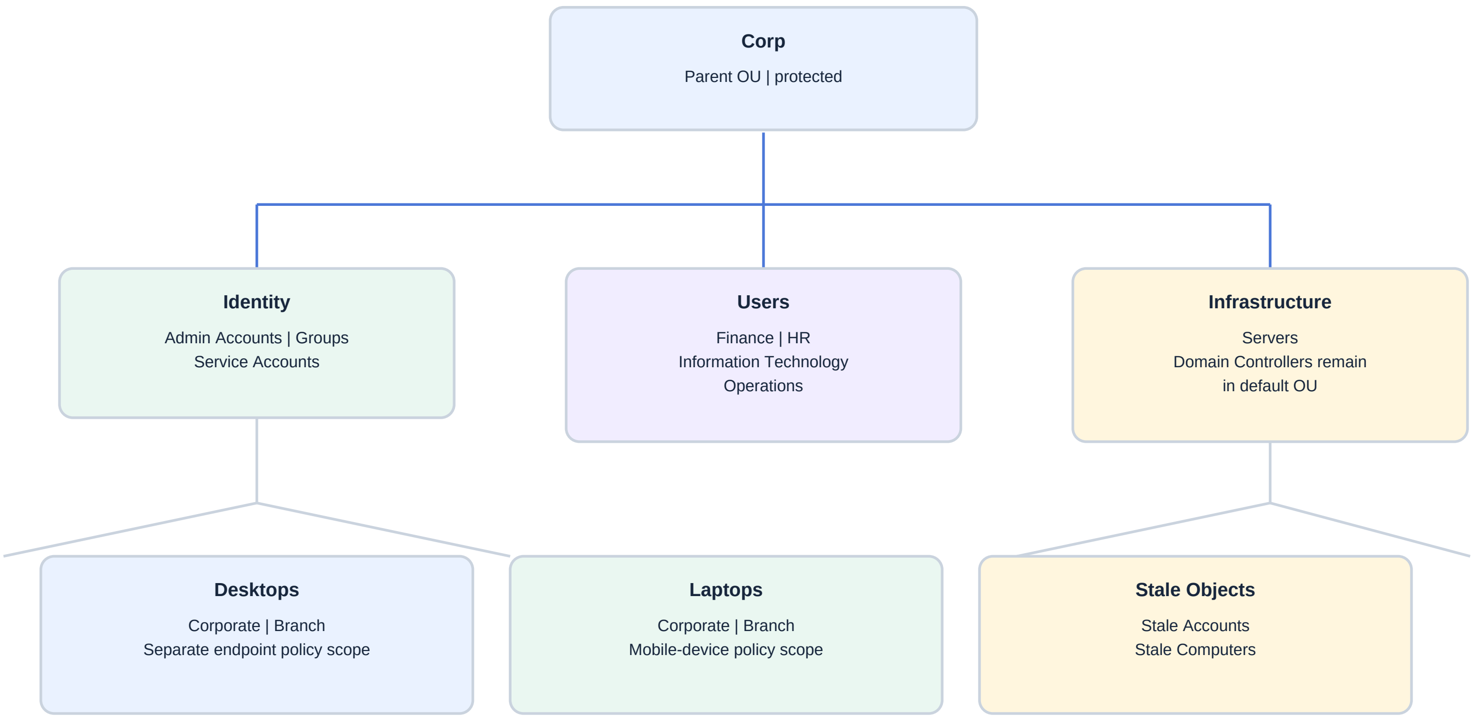


Figure 1 - Functional OU hierarchy implemented beneath the Corp parent OU.

Implemented Hierarchy

Parent	Child OU(s)	Administrative Purpose
Corp	Admin Accounts	Privileged identities
Corp	Groups	Security and distribution groups
Corp	Servers	Member-server computer objects
Corp	Service Accounts	Application and service identities
Corp	Stale Accounts / Stale Computers	Lifecycle quarantine and review
Corp > Users	Finance / HR / Information Technology / Operations	Department policy and delegation
Corp > Desktops	Corporate / Branch	Fixed endpoints by location
Corp > Laptops	Corporate / Branch	Mobile endpoints by location

Implementation Summary

1. Used a pre-change checkpoint approach before modifying the directory; the supplied screenshot shows the earlier p02-PreADDS checkpoint rather than a Project 03-specific label.
2. Opened Active Directory Users and Computers and enabled Advanced Features.
3. Created the Corp parent OU beneath corp.mslab.test with accidental-deletion protection enabled.
4. Created Admin Accounts, Groups, Servers, Service Accounts, Stale Accounts, Stale Computers, and Users beneath Corp.
5. Created separate Desktops and Laptops OUs instead of a single Workstations container.
6. Created Corporate and Branch child OUs beneath both Desktops and Laptops.
7. Created Finance, HR, Information Technology, and Operations beneath Users.
8. Left CORP-DC1 in the default Domain Controllers OU to preserve domain-controller policy scope.
9. Queried the hierarchy with PowerShell and confirmed every displayed descendant OU was protected from accidental deletion.
10. Opened the Operations OU Object tab and independently verified the protection setting was enabled.

Design Decisions

Decision	Operational Value
Stale Accounts and Stale Computers	Separates disabled identities from retired devices for review, reporting, and retention
Desktops and Laptops	Supports distinct security baselines, BitLocker, power, update, and mobility policies
Corporate and Branch	Prepares each endpoint class for location-specific configuration and administration
Departmental Users	Creates future scope for delegation and business-specific policy
Default Domain Controllers OU	Keeps CORP-DC1 within the domain-controller-specific policy boundary

SCOPE

Project 03 creates only the directory structure and its protection controls. Object creation, group strategy, delegation, domain joins, and Group Policy remain separate portfolio projects.

Evidence - Available Hyper-V Recovery Point

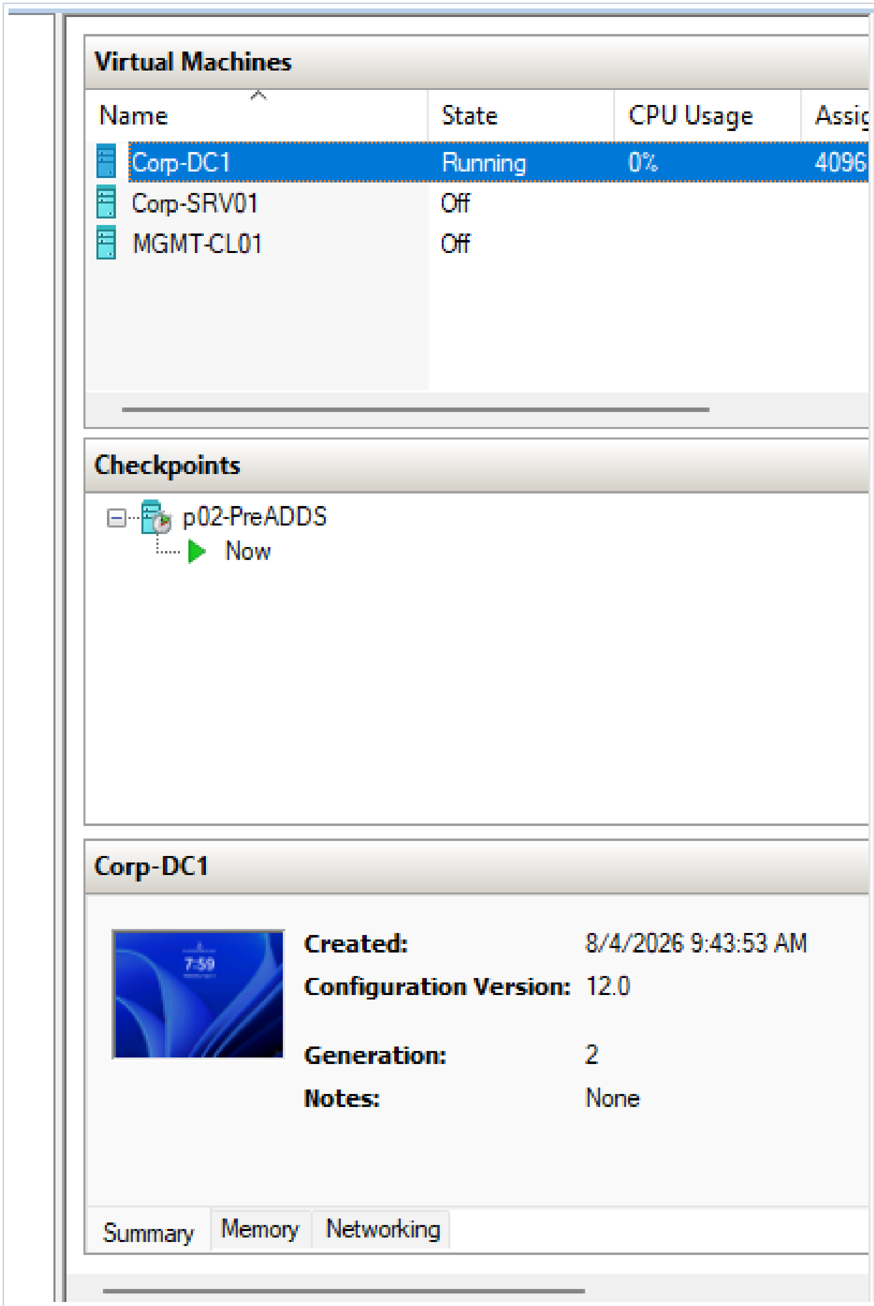


Figure 2 - Existing CORP-DC1 Hyper-V Checkpoint

Hyper-V Manager shows p02-PreADDS on CORP-DC1. This is an existing recovery point, but the screenshot does not independently verify a Project 03-specific checkpoint.

Change-Control Rationale

The user reported creating a checkpoint before modifying the directory structure. The supplied screenshot, however, shows p02-PreADDS, which predates the Active Directory deployment and is not a clean OU-only rollback point. A current Project 03 checkpoint screenshot would strengthen the change-control record. Production domain-controller recovery should use supported system-state backup and forest-recovery procedures.

CONTROL	The screenshot is retained for transparency, but its p02-PreADDS label makes Project 03 checkpoint evidence partial. It is not a replacement for Active Directory backup, replication, or tested recovery procedures.

Evidence - Implemented Organizational Unit Hierarchy

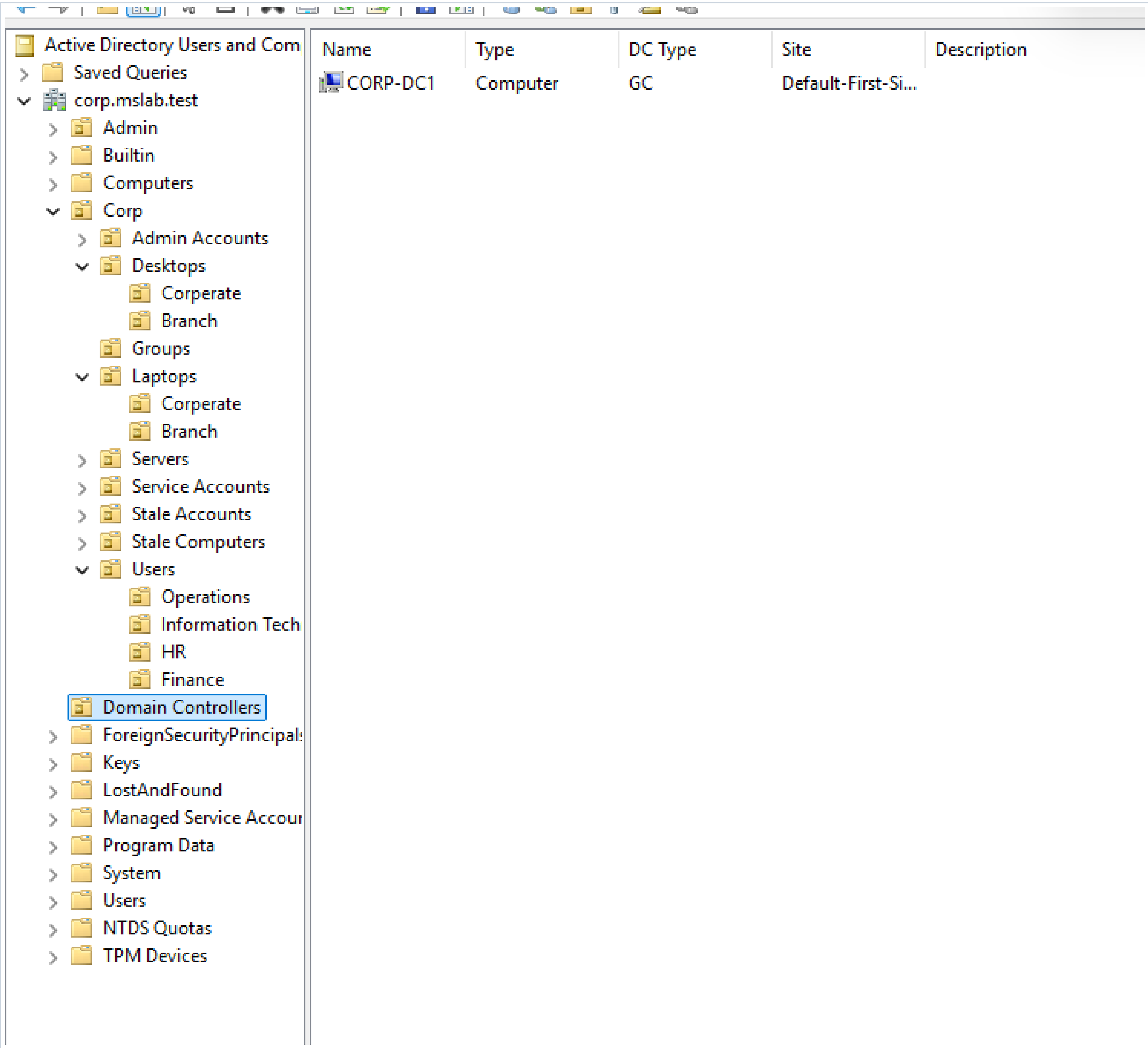
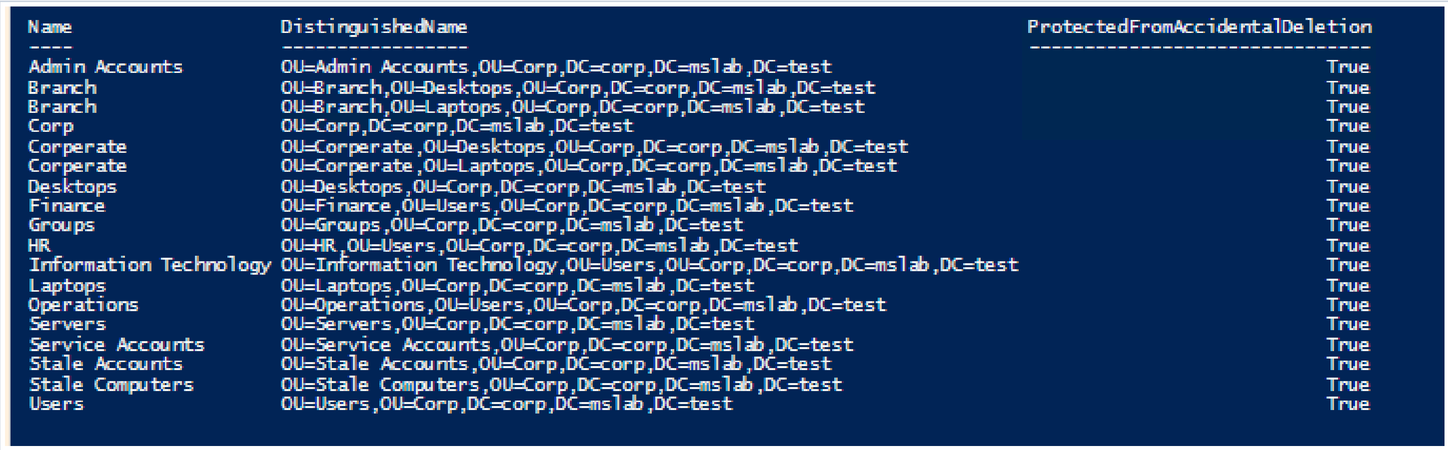


Figure 3 - Completed OU Structure in Active Directory Users and Computers

The Corp hierarchy separates privileged accounts, endpoint classes and locations, groups, servers, service accounts, stale objects, and departmental users. CORP-DC1 remains in the default Domain Controllers OU.

DESIGN	Separating laptops from desktops and stale accounts from stale computers improves future policy targeting and lifecycle administration compared with broader Workstations and Disabled Objects containers.

Evidence - PowerShell OU Protection Validation



```
Name                   DistinguishedName                                                   ProtectedFromAccidentalDeletion
----                   -----------------                                                   -------------------------------
Admin Accounts         OU=Admin Accounts,OU=Corp,DC=corp,DC=mslab,DC=test                                             True
Branch                 OU=Branch,OU=Desktops,OU=Corp,DC=corp,DC=mslab,DC=test                                         True
Branch                 OU=Branch,OU=Laptops,OU=Corp,DC=corp,DC=mslab,DC=test                                          True
Corp                   OU=Corp,DC=corp,DC=mslab,DC=test                                                               True
Corperate              OU=Corperate,OU=Desktops,OU=Corp,DC=corp,DC=mslab,DC=test                                      True
Corperate              OU=Corperate,OU=Laptops,OU=Corp,DC=corp,DC=mslab,DC=test                                       True
Desktops               OU=Desktops,OU=Corp,DC=corp,DC=mslab,DC=test                                                   True
Finance                OU=Finance,OU=Users,OU=Corp,DC=corp,DC=mslab,DC=test                                           True
Groups                 OU=Groups,OU=Corp,DC=corp,DC=mslab,DC=test                                                     True
HR                     OU=HR,OU=Users,OU=Corp,DC=corp,DC=mslab,DC=test                                                True
Information Technology OU=Information Technology,OU=Users,OU=Corp,DC=corp,DC=mslab,DC=test                           True
Laptops                OU=Laptops,OU=Corp,DC=corp,DC=mslab,DC=test                                                    True
Operations             OU=Operations,OU=Users,OU=Corp,DC=corp,DC=mslab,DC=test                                        True
Servers                OU=Servers,OU=Corp,DC=corp,DC=mslab,DC=test                                                    True
Service Accounts       OU=Service Accounts,OU=Corp,DC=corp,DC=mslab,DC=test                                           True
Stale Accounts         OU=Stale Accounts,OU=Corp,DC=corp,DC=mslab,DC=test                                             True
Stale Computers        OU=Stale Computers,OU=Corp,DC=corp,DC=mslab,DC=test                                            True
Users                  OU=Users,OU=Corp,DC=corp,DC=mslab,DC=test                                                      True
```

Figure 4 - Descendant OU Distinguished Names and Protection State

PowerShell lists 17 OUs beneath the Corp search base and shows ProtectedFromAccidentalDeletion as True for every displayed object.

Validation Interpretation

Check	Result	Evidence Meaning
Custom OU design	18 total	Corp parent plus 17 descendants
Displayed query results	17 descendants	Get-ADOrganizationalUnit search beneath OU=Corp
Displayed protection state	17 of 17 True	No unprotected descendant OU appears in the evidence
Distinguished-name placement	PASS	Objects are nested beneath the intended parent paths

Important Count Detail

The PowerShell screenshot displays the 17 descendant OUs beneath **OU=Corp**. The complete design contains 18 custom OUs when the Corp parent itself is included. This distinction keeps the validation claim aligned with the command output shown in the evidence.

VALIDATED	Every OU returned by the descendant query is protected from accidental deletion, and each distinguished name matches the intended hierarchy.

Evidence - Accidental-Deletion Protection

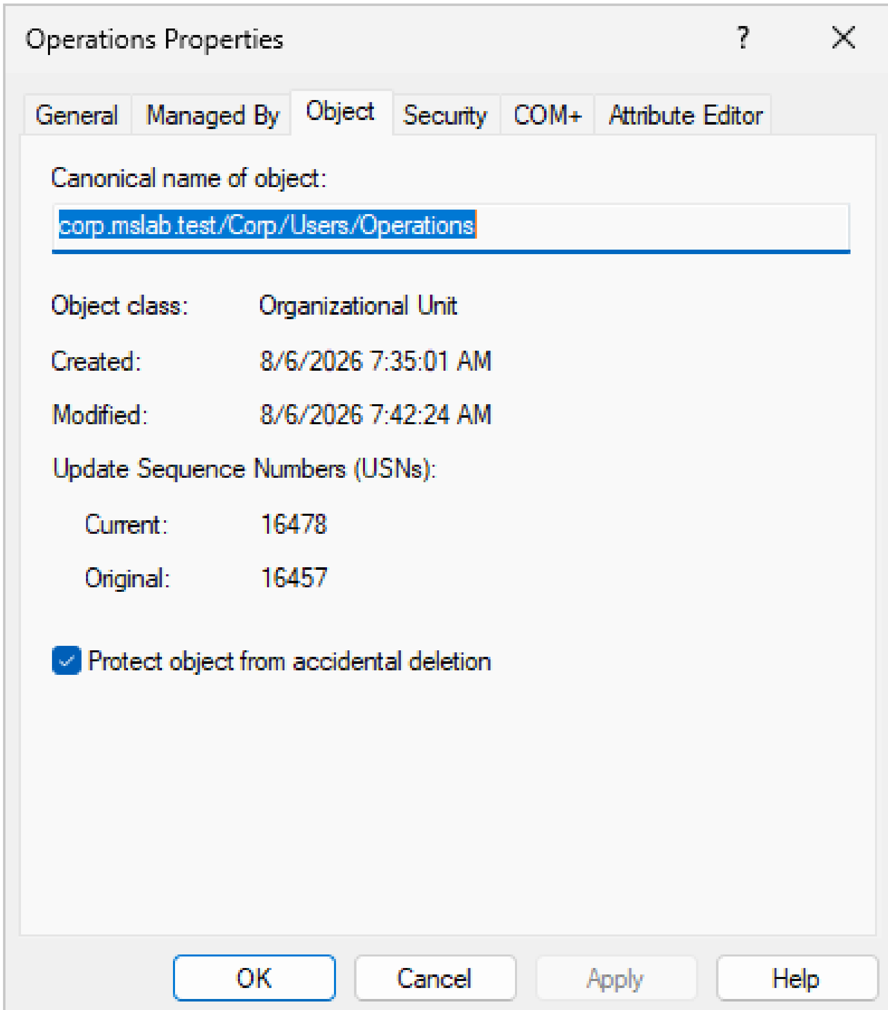


Figure 5 - Operations OU Object Protection

The Object tab for corp.mslab.test/Corp/Users/Operations shows Protect object from accidental deletion enabled.

Control Purpose

Accidental-deletion protection adds deny-delete permissions that help prevent administrators from removing an OU and its contents unintentionally. It does not prevent an authorized administrator from deliberately removing the protection setting, so change control, least privilege, auditing, and backup remain necessary.

DEFENSE	The GUI confirmation supports the PowerShell evidence with an independent view of the Operations OU protection setting.

Policy and Delegation Readiness

OU Scope	Future Control	Potential Delegation
Admin Accounts	Privileged-access controls, stronger authentication, restricted logon	Privileged identity administrators
Users by department	User settings and department-specific configuration	Department or service-desk operators
Desktops by location	Fixed-device baselines, software, firewall, and update policy	Endpoint administrators
Laptops by location	BitLocker, mobility, VPN, power, and remote-management policy	Endpoint administrators
Servers	Server security baseline, update rings, firewall, and audit policy	Server administrators
Service Accounts	Logon restrictions, auditing, and managed-service-account transition	Identity administrators
Stale objects	Restricted policy, review, retention, and controlled deletion	Service desk with approval workflow

Administrative Model

The hierarchy is organized around stable administrative boundaries rather than the current organization chart alone. Object class, device type, location, privilege level, and lifecycle status are separated where they are likely to require different policy or delegated ownership. This reduces the need to redesign the directory when users and computers are onboarded.

Naming and Placement Standards

Standard	Application
Clear names	Human-readable OU labels reflect the object purpose
Consistent location labels	Corporate and Branch are used beneath both endpoint classes
Dedicated stale containers	Objects can be disabled and reviewed before deletion
Protected structure	Deletion protection reduces unintended structural changes
DC placement	Domain controllers remain in the built-in Domain Controllers OU

Validation Results

Validation	Result	Evidence
Project 03-specific checkpoint	PARTIAL	Figure 2 shows p02-PreADDS; creation was reported but not independently shown
Corp parent and purpose-built child OUs created	PASS	Figure 3
Desktops and Laptops separated by Corporate and Branch	PASS	Figure 3
Users separated by Finance, HR, IT, and Operations	PASS	Figure 3
Stale accounts and computers separated	PASS	Figure 3
17 descendant OUs returned with protection set to True	PASS	Figure 4
Operations OU protection independently confirmed	PASS	Figure 5
CORP-DC1 retained in Domain Controllers OU	PASS	Figure 3

Security Analysis

OU design is not itself a security boundary, but it is the control plane for Group Policy scope and delegated administration. Separating privileged identities, service accounts, user departments, endpoint types, server objects, and stale objects reduces accidental cross-scope policy changes and prepares the domain for least-privilege delegation.

Accidental-deletion protection reduces the risk of unintended removal. A correctly labeled, current pre-change checkpoint would improve short-term lab recoverability, but neither control replaces backups, change approval, auditing, or a second domain controller.

Evidence Limitations

The evidence confirms the implemented hierarchy, 17 descendant distinguished names with protection set to True, and one GUI protection setting. The Hyper-V screenshot shows p02-PreADDS rather than a Project 03-specific checkpoint, so that checkpoint claim is only partially evidenced. It also does not include a separate PowerShell row for the Corp parent, delegated permissions, linked Group Policy Objects, audit events, backups, or object-population tests. Those items remain appropriate evidence for later projects.

Lessons Learned

- OU structure should follow administrative and policy boundaries rather than mirror every organizational detail.
- Separating laptops and desktops creates useful policy scope for mobility, encryption, power, and update controls.
- Separate stale-account and stale-computer OUs support different review, retention, and deletion workflows.
- Domain controllers should remain in the built-in Domain Controllers OU unless a carefully designed alternative is required.
- Distinguished-name and protection queries provide stronger validation than relying only on the ADUC tree view.

Production Improvements

- Define role-based groups and delegate OU administration through least-privilege access-control entries.
- Link tested security baselines and operational GPOs at the intended scopes.
- Establish naming, ownership, retention, and approval standards for every OU.
- Enable advanced auditing and centralized event collection for directory changes.
- Implement system-state backups, a second domain controller, and tested recovery procedures.
- Add scripted compliance checks for placement, protection, empty containers, and stale-object age.

Skills Demonstrated

Active Directory	Governance / Security	Validation / Operations
OU hierarchy design	Accidental-deletion protection	PowerShell AD queries
Distinguished-name analysis	Policy-scope planning	Evidence-based validation
Object-lifecycle separation	Least-privilege readiness	Checkpoint change control
Endpoint segmentation	Administrative boundary design	ADUC advanced features

Resume Bullet

RESUME	Designed and validated an 18-OU Active Directory hierarchy for a Windows Server 2025 enterprise lab, segmenting privileged identities, departments, servers, service accounts, endpoint classes and locations, and stale objects while enforcing accidental-deletion protection and verifying placement with PowerShell.

STAR Interview Story

Situation: The new corp.mslab.test forest was operational, but custom users, computers, groups, and servers would have entered a flat structure without consistent policy or administrative scope.

Task: Build a scalable OU hierarchy before onboarding domain objects and add safeguards against accidental structural deletion.

Action: I used a pre-change recovery approach, implemented a protected Corp hierarchy, separated identities, departments, servers, service accounts, endpoint types and locations, and stale objects, retained the domain controller in its default OU, and validated distinguished names and deletion protection with PowerShell and ADUC.

Result: The domain now has an 18-OU administrative framework ready for controlled object creation, least-privilege delegation, Group Policy targeting, lifecycle management, and future corporate and branch onboarding.

Next Lab Handoff

Project 04 can populate the directory with users, administrative accounts, service identities, and role-based security groups. It should document naming standards, group scope and type, membership strategy, account placement, disabled-object handling, and PowerShell validation before member systems are domain joined.

Version History

Version	Release Date	Description
v1.0.0	August 6, 2026	Initial documented implementation. Created the protected Corp hierarchy and 17 descendant OUs, separated identities, departments, servers, endpoint types, locations, and stale objects, validated OU paths and protection, documented the supplied checkpoint-label limitation, and embedded all supplied evidence using portfolio documentation standard v1.0.1.

References

- Microsoft Learn - Active Directory Domain Services overview
- Microsoft Learn - Designing an OU structure that supports Group Policy
- Microsoft Learn - Get-ADOrganizationalUnit PowerShell reference
- Microsoft Learn - Protecting objects from accidental deletion
- User-provided Hyper-V checkpoint, ADUC hierarchy, PowerShell OU, and Object-tab evidence